

Frontend Unit Testing Documentation

Plant Disease Detection System – Frontend Module

This document presents unit testing performed on the frontend module using Flutter's testing framework. Each test validates individual components and services to ensure correctness and reliability.

Tools Used:

Flutter Test Framework

Dart

Tests:

1. Diagnosis API Test

Verified that the diagnosis service correctly handles API responses and exceptions using a mocked HTTP client.

Command: flutter test test/diagnosis_api_test.dart

```
PS C:\Users\DELL\Downloads\Pl flutter test test/diagnosis_api_test.dart detection-main\frontend>
00:01 +2: (tearDownAll)

All tests passed!
```

2. Weather API Test

Validated that weather data is correctly fetched and parsed from the external API, ensuring robust error handling.

Command: flutter test test/weather_api_test.dart

```
PS C:\Users\DELL\Downloads\Pl flutter test test/weather_api_test.dart e_detection-main\frontend>
00:01 +2: (tearDownAll)

All tests passed!
```

3. Token Storage Test

Confirmed that authentication tokens and user history are securely stored, retrieved, and deleted handling local storage.

Command: flutter test test/token_storage_test.dart

```
PS C:\Users\DELL\Downloads\Plant_disease_detection-main\Plant_disease_detection-main\frontend> flutter test test/token_storage_test.dart
00:01 +3: (tearDownAll)
_api_test.dart
00:01 +2: (tearDownAll)
```

4. Auth API Test

Tests user registration and login flows, ensuring that credentials are sent correctly and tokens are received.

Command: flutter test test/auth_api_test.dart

```
PS C:\Users\DELL\Downloads\Plant_disease_detection-main\Plant_disease_detection-main\frontend> flutter test test/auth
_api_test.dart
00:01 +2: (tearDownAll)

All tests passed!
00:01 +2: All tests passed!
```

5. History API Test

Verifies that the app correctly fetches and deserializes the user's diagnosis history list from the backend.

Command: flutter test test/history_api_test.dart

```
PS C:\Users\DELL\Downloads\Plant_disease_detection-main\Plant_disease_detection-main\frontend> flutter test test/history_api_test.dart
00:01 +2: (tearDownAll)

All tests passed!
```

6. User API Test

Verified that user profile data can be fetched and updated (PATCH requests) correctly.

Command: flutter test test/user_api_test.dart

```
PS C:\Users\DELL\Downloads\Plant_disease_detection-main\Plant_disease_detection-main\frontend> flutter test test/user_api_test.dart
00:01 +2: (tearDownAll)

All tests passed!
```

7. Notification API Test

Tested fetching notification lists and marking them as read via the API.

Command: flutter test test/notification_api_test.dart

```
PS C:\Users\DELL\Downloads\Plant_disease_detection-main\Plant_disease_detection-main\frontend> flutter test test/notification_api_test.dart
00:01 +2: (tearDownAll)
All tests passed!
```

Conclusion:

All frontend unit tests executed successfully. This confirms that the tested components function as expected under defined conditions.